



SOMAIYA
VIDYAVIHAR UNIVERSITY
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Deployment Strategies in a Kubernetes - Based Microservices Architecture



Department of Information Technology



A Comparative Performance Analysis of Canary vs. Blue - Green Deployment Strategies

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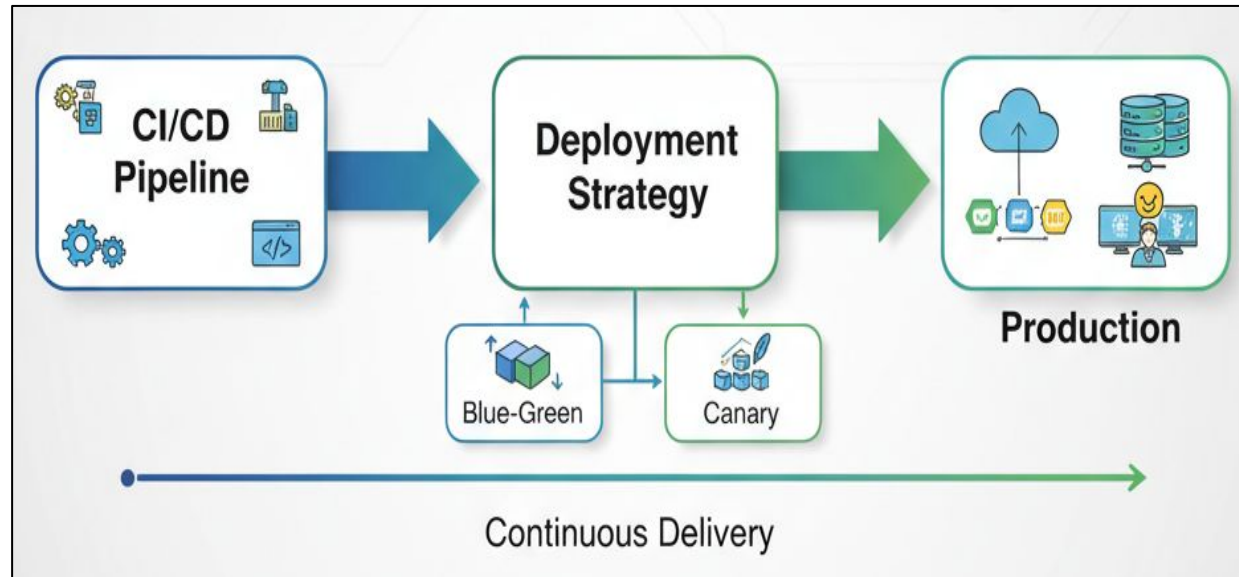
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The Challenge of Zero - Downtime Continuous Delivery (CD)

- **The DevOps Mandate:** Organizations strive for rapid and reliable software releases by enhancing CI/CD pipelines.
- **Core Problem:** How to deploy new application versions while minimizing downtime and mitigating risks?
- **Blue-Green Deployment (BG):** Involves maintaining two identical production environments (live and idle).
- **Canary Deployment:** Gradually rolls out new versions to a subset of users, enabling early issue detection.



Strategy 1: Blue-Green Deployment (The "Instant Switch")

The live system (Blue) and the new version (Green) run in parallel; traffic is switched between them.

Key Advantages (Speed & Recovery)

- Enables seamless rollbacks.
- Provides minimal downtime during the switch.

Key Disadvantages (Cost & Risk)

- High Infrastructure Overhead: Requires duplication of the production environment.
- Infrastructure overhead was reported at 65%.

Strategy 2: Canary Deployment (The "Gradual Rollout")

New release is exposed to a small subset of users before a full rollout. This allows for real-time monitoring and validation.

Key Advantages (Risk Mitigation)

- Exhibited a lower failure rate (3%) than BG (5%).
- Perceived to be more effective (85%) in reducing risk exposure.
- Showed slightly better system stability (99.98% uptime).

Key Disadvantages (Complexity)

- Longer Time-to-Deploy (30 mins) compared to BG (15 mins).
- Requires sophisticated traffic routing complexities and monitoring overhead.

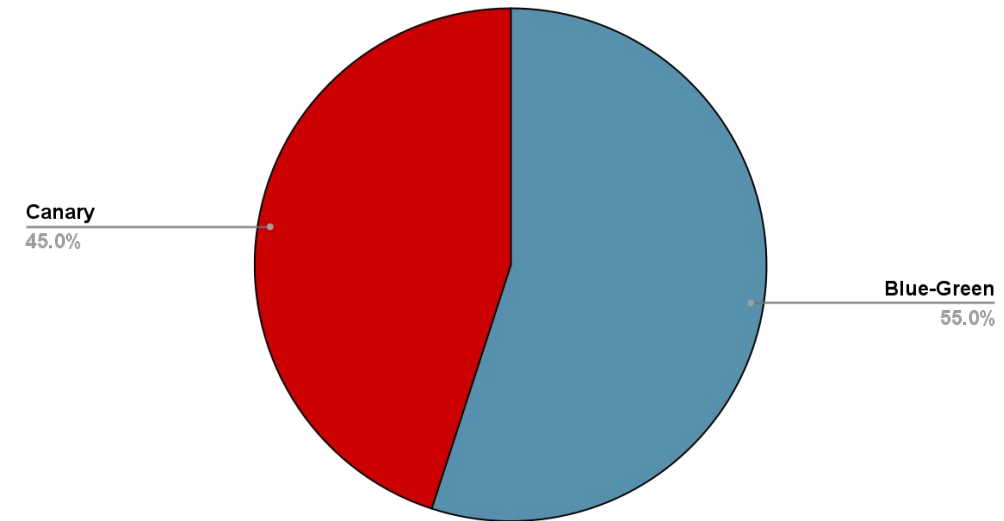
Quantitative Analysis: Comparing Core Performance Metrics

Metric	Blue-Green Deployment	Canary Deployment
Average Deployment Time (TTD)	15 mins	30 mins
Failure Rate	5%	3%
User Impact (Errors/1000 users)	12	5
Infrastructure Overhead	65%	50%

Statistical Analysis: The Trade-Offs (Risk vs. Cost)

- **Failure Rate (Canary Wins):** T-Test confirmed Canary Deployments significantly reduce failure rates ($p < 0.05$). Fewer user disruptions (5 vs. 12 error reports per 1000 users).
- **Rollback Success (Comparable):** Chi-Square Test showed no significant difference in rollback success rates ($p > 0.05$). Both provide robust failure recovery.
- **Infrastructure Cost (BG Loses):** ANOVA confirmed Blue-Green requires significantly higher infrastructure investment ($p < 0.01$).

Infrastructure Overhead



Industry Adoption and the Rise of the Hybrid Model

Case Studies: Strategy Alignment



Netflix (Canary): Prioritizes low risk/user impact; uses AI-driven anomaly detection.

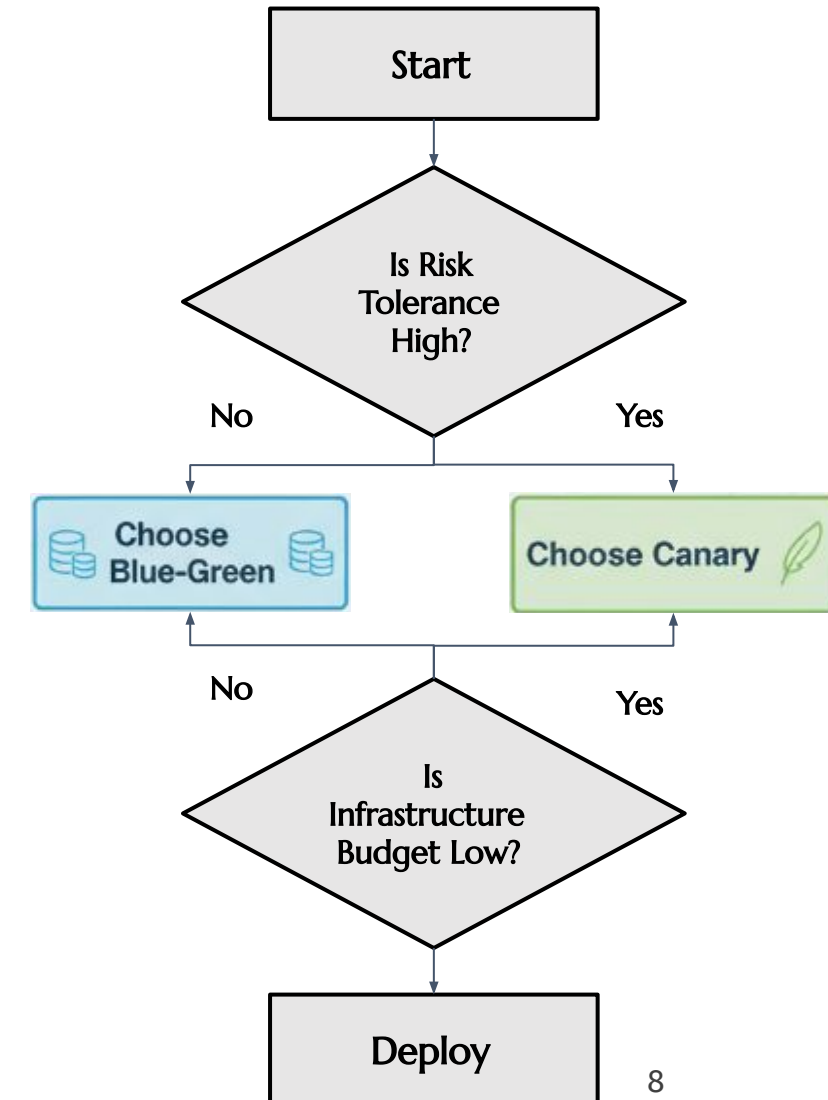
Amazon (Blue-Green): Prioritizes instant rollback; critical for transactional security (e-commerce).



The Hybrid Model (Google):

Combines Canary for new features and Blue-Green for critical system updates.

Goal: Optimize both speed and stability.



Conclusion: The Strategic Deployment Choice

Summary of Trade-Offs:

- **Blue-Green:** Best for Speed (15 mins TTD) and Instant Rollback. High resource cost.
- **Canary:** Best for Risk Mitigation (3% failure) and Stability. Longer rollout time.

Key Recommendations:

- **Strategic Alignment:** Match strategy to the application's risk profile (e.g., Canary for financial, BG for SaaS).
- **Cost:** Small-to-medium enterprises (SMEs) should favor Canary to avoid high BG infrastructure costs.
- **Future:** Adopt a Hybrid Model and invest in AI-driven monitoring for optimal outcomes.

References

1. M. Idowu and B. Barnty, "A Deep Dive into Blue-Green and Canary Deployments: Benefits, Challenges, and Best Practices," ResearchGate, Article, March 2025. [Online].

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<https://www.researchgate.net/publication/390108683>

A Deep Dive into Blue-Green and Canary Deployments: Benefits, Challenges, and Best Practices

Marvel Idowu

Abstract

In modern software development, **blue-green deployments** and **canary deployments** are widely adopted strategies for minimizing downtime and mitigating risks during application releases. **Blue-green deployment** involves maintaining two production environments—one live and one idle—allowing seamless rollback and reducing deployment failures. In contrast, **canary deployment** gradually rolls out new versions to a subset of users, enabling early detection of issues before full release. This paper explores the **benefits, challenges, and best practices** associated with both deployment strategies. Key benefits include **reduced downtime, improved rollback mechanisms, and enhanced user experience**, while challenges such as **infrastructure overhead, traffic routing complexities, and monitoring requirements** are also examined. Best practices such as **automated testing, robust monitoring, traffic segmentation, and gradual rollout strategies** are discussed to ensure successful deployment. By understanding these strategies, organizations can optimize their **DevOps and CI/CD pipelines**, enhancing **continuous delivery and deployment** in modern software ecosystems.

Keywords: Blue-Green Deployment, Canary Deployment, CI/CD, DevOps, Continuous Delivery, Continuous Deployment, Rollback Strategies, Traffic Routing, Automated Testing, Monitoring